What is Muse data?

MUSE stands for **Multi-Unit Spectroscopic Explorer**. It's an instrument on the **Very Large Telescope (VLT)** in Chile, operated by ESO (European Southern Observatory).

it's an **Integral Field Unit (IFU)**

A normal camera takes a 2D image. A normal spectrograph takes a spectrum (light split by wavelength) of one point. MUSE does **both simultaneously** — it takes a spectrum at **every single pixel** across a large field of view (1 arcminute × 1 arcminute). So instead of a single image or a single spectrum, you get a **3D data cube**:

- X position
- Y position
- Wavelength (i.e., the spectrum at each position)

This is incredibly powerful because you can extract images at any wavelength or extract a spectrum from any location in the field.

Key MUSE specs:

- Wavelength range: **4000–9000 Å** (optical light)
- ~3000 resolution elements across the field
- Each element gives ~100k pixels worth of spectral information
- This matches exactly what your project description says!

What is fits file format?

FITS stands for **Flexible Image Transport System**. It is the standard file format used in astronomy , almost all astronomical data, including MUSE data cubes, is stored as `.fits` files.

A FITS file can contain:

- **Images** (2D arrays of pixel values)
- **Data cubes** (3D arrays — like MUSE data: x, y, wavelength)
- **Tables** (catalogues of stars, measurements, etc.)
- **Headers** — metadata stored as plain text, containing information like the telescope used, observation date, coordinates, exposure time, wavelength calibration, etc.

For my project, the MUSE data cube will likely be a FITS file with shape something like **(wavelength channels × y pixels × x pixels)**

Research Paper-1(The Th 28 Jet)

What is this paper about?

Th 28 (also called Sz 102 or Krautter's Star) is a **Classical T Tauri Star** — a young star still in its pre-main-sequence phase, located in the **Lupus 3 cloud** about 160 parsecs away. It drives a **bipolar jet** (gas shooting out in two opposite directions), called HH 228.

Why are jets interesting? Jets from young stars are linked to how stars form — they remove mass and angular momentum from the star-disk system. Understanding jets helps us understand how stars like our Sun were born.

What did the researchers do with MUSE data?

They pointed MUSE at Th 28 and used its IFU capability to study the jet in detail:

1. **Spectro-imaging** — They extracted images of the jet in specific emission lines (like Hα, [S II], [O III], [N II]) and at specific velocity ranges. This lets them see which parts of the jet are moving towards us (blue-shifted) vs. away (red-shifted).
2. **Position-Velocity (PV) diagrams** — Slices through the data cube showing how velocity changes with position along the jet.
3. **Knot identification** — They found **6 new knots** (clumps of shocked gas) in the jet that hadn't been seen before, including some only visible after applying a **Lucy-Richardson deconvolution** to improve spatial resolution.
4. **Jet width measurement** — By fitting Gaussian profiles across the jet, they measured how wide it is and its opening angle.
5. **Precession detection** — Most importantly, they found the jet **wiggles** with **point symmetry** on both sides of the star. This is a signature of **jet precession** — the jet axis is slowly rotating like a spinning top.

What are they searching for?

The core scientific question is: **why does the jet precess?** Their answer: probably a **brown dwarf companion** (a failed star, ~70 Jupiter masses) orbiting Th 28 at a very close separation of ~0.3 AU, causing the inner disk to precess, which makes the jet precess. This has a period of about **8 years**.

This connects back to the bigger question of **jet launching** — how exactly do young stars produce these jets, and what does the jet's shape tell us about the star-disk system?

Research Paper - 2(AION-1 (Parker et al. 2025)

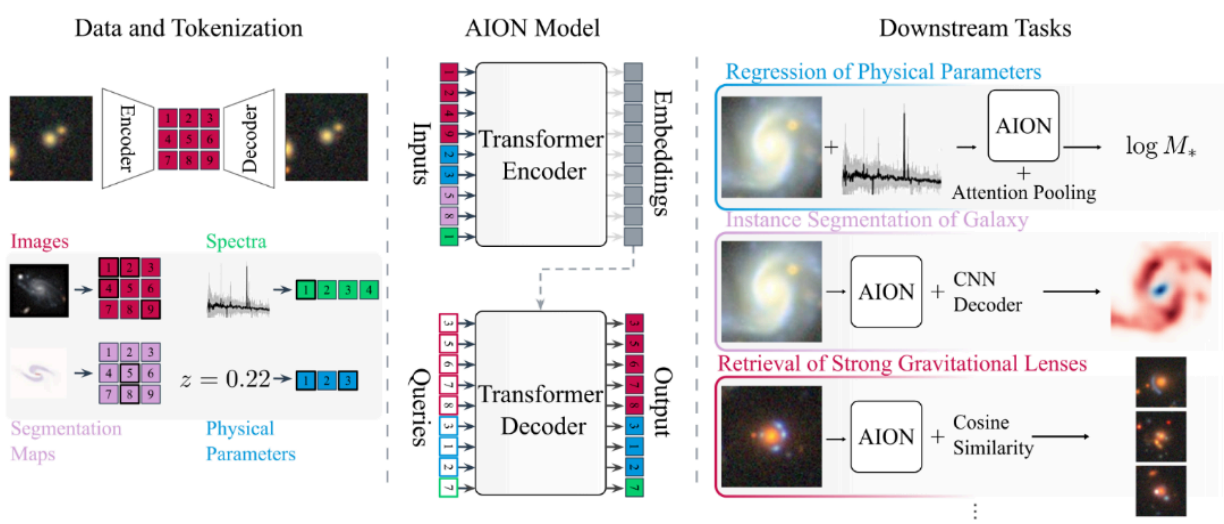
AION-1 is a **foundation model** (think: a large AI model like GPT, but for astronomy data) that can process 39 different types of astronomical data — images from multiple telescopes, spectra, and scalar measurements — all in a single unified model.

AION-1 addresses a major limitation in modern astronomy: the absence of a unified framework capable of jointly modelling heterogeneous data modalities. Traditional analyses rely on separate, instrument-specific models, preventing consistent cross-modal reasoning. AION-1 overcomes this by introducing an omnimodal architecture that integrates images, spectra, and scalar measurements within a single system.

1.1 Architecture and Methodology

AION-1 employs a two-stage design tailored to the diversity and noise characteristics of real astronomical observations.

- **Universal Tokenization:** All input modalities—2D images, 1D spectra, and scalar values—are transformed into a common discrete token space.
  - **Images:** Encoded using a ResNet-based MagViT autoencoder with Finite Scale Quantization (FSQ).
  - **Spectra:** Processed through a ConvNeXt-V2 backbone using Look-up-Free Quantization (LFQ).
  - **Scalars:** Normalized via CDF-mapping to a standard normal distribution before FSQ quantization.
- **Multimodal Masked Modeling:** A transformer encoder–decoder is trained with a masked-token objective, enabling the model to learn both intra-modal and cross-modal relationships. By reconstructing masked tokens from partially observed inputs, AION-1 develops a unified physical understanding across all data types.



Three Model variants of AION-1

Model Variant	Parameters	Encoder Blocks	Decoder Blocks	MLP Dimension
AION-B	300M	12	12	3072
AION-L	800M	24	24	4096
AION-XL	3.1B	24	24	16384

What is precessing??

What is red and blue shift?? And how is it used to identify objects from image(via wavelength)

What is flux?